Waipapa Taumata Rau University of Auckland

What is Deep Learning

SOFTENG 753: Machine Learning Techniques and Applications

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Content

- The AI / ML / DL Landscape
- Core Concepts: Representations & Data
- Training Loop in DL
- Current State and Modern Era

Artificial Intelligence

- Artificial Intelligence (AI) can be described as *the effort to automate intellectual tasks normally performed by humans.*

- Artificial intelligence crystallized as a field of research in 1956, when John McCarthy, then a young Assistant Professor of Mathematics at Dartmouth College, organized a summer workshop under the following proposal:

  > *The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.*

- At the end of the summer, the workshop concluded without having fully solved the riddle it set out to investigate.
  - It was attended by many people who would move on to become pioneers in the field, and it set in motion an intellectual revolution that is still ongoing to this day.

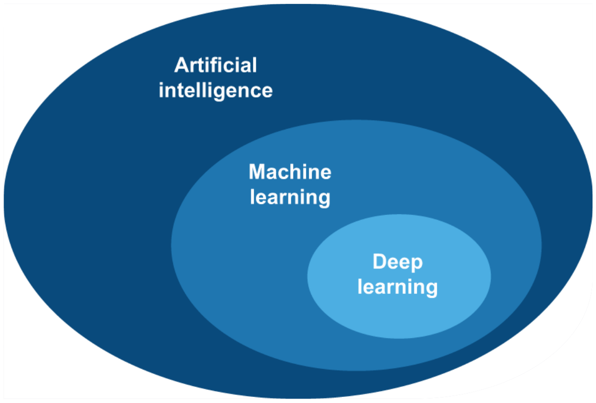
Symbolic AI

- Most experts believed that human-level artificial intelligence could be achieved by having programmers handcraft a sufficiently large set of explicit rules for manipulating knowledge stored in explicit databases.
- **Symbolic AI** was the dominant paradigm in AI from the 1950s to the late 1980s.
- Proved suitable to solve well-defined, logical problems, such as playing chess.
- Reached its peak popularity during the expert systems boom of the 1980s.
- It was intractable to figure out explicit rules for solving more complex, fuzzy problems, such as image classification, speech recognition, or natural language translation.

Artificial Intelligence, Machine Learning, and Deep Learning

- **AI:** Any technique that enables machines to mimic human behavior
- **ML:** Systems that learn from data rather than following explicit rules
- **DL:** Multi-layer neural networks that learn hierarchical representations

AI is a general field that encompasses machine learning and deep learning, but that also includes many more approaches that may not involve any learning.



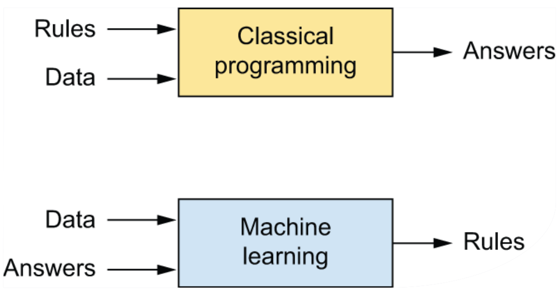
Early chess programs, which only involved hardcoded rules crafted by programmers didn't qualify as machine learning.

Machine Learning

- A machine learning system is *trained* rather than explicitly programmed.
- It is presented with many examples relevant to a task.
- It finds statistical structure in these examples that eventually allows the system to come up with rules for automating the task.
- Machine learning started to flourish in the 1990s and has quickly become the most popular and most successful subfield of AI.
  - Driven by the availability of faster hardware and larger datasets.

The usual way to make a computer do useful work is to have a human programmer write down rules as a computer program.

The program is followed to turn input data into appropriate answers.



Machine learning turns around the approach.

The machine looks at the input data and the corresponding answers and figures out what the rules should be.

Learning Rules and Representations from Data

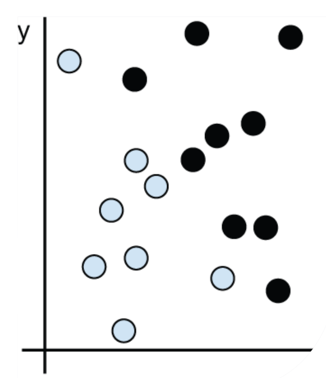
Machine Learning discovers rules to execute a data processing task

- Machine Learning requires:
  - **Input data points**
    - if the task is speech recognition, these data points could be sound files of people speaking.
    - If the task is image tagging, they could be pictures.
  - **Examples of the expected output**
    - In a speech-recognition task, these could be human-generated transcripts of sound files.
    - In an image task, expected outputs could be tags such as “dog,” “cat,” and so on.
  - **A way to measure whether the algorithm is doing a good job**
    - This is necessary to determine the distance between the algorithm’s current output and its expected output.
    - The measurement is used as a feedback signal to adjust the way the algorithm works. This adjustment step is what we call **learning**.

Learning Rules and Representations from Data

Machine learning models are all about finding appropriate representations for their input data

- The central problem in machine learning and deep learning is to meaningfully transform data to learn useful representations of the input data
  - Representations that get us closer to the expected output.
- A color image can be encoded in the RGB format (red-green-blue) or in the HSV format (hue-saturation-value)
  - “Select all red pixels in the image” is simpler in the RGB format.
  - “Make the image less saturated” is simpler in the HSV format.
- Develop an algorithm that can take a point’s (x, y) coordinates and output whether that point is likely black or white.
  - The inputs are the coordinates of our points.
  - The expected outputs are the colors of our points.
  - A way to measure whether our algorithm is doing a good job could be the percentage of points that are being correctly classified.

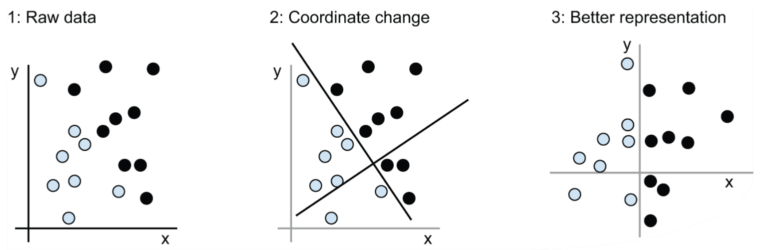


Develop an algorithm that can take a point’s (x, y) coordinates and output whether that point is likely black or white.

Data Representation

- Finding useful representations by hand is a hard work.
- Learning, in the context of machine learning, describes an automatic search process for data transformations that produce useful representations of some data, guided by some feedback signal.
- Machine learning algorithms aren't usually creative in finding these transformations; they're merely searching through a predefined set of operations, called a hypothesis space.
  - The space of all possible coordinate changes is our hypothesis space in this example.

We need a new representation of our data that cleanly separates the white points from the black points.



Raw data forms a highly folded, complex manifold (like a crumpled paper ball).

A representations is a different way to look at data. We could use a coordinate change.

The new representation cleanly separates the data points, reducing a complex classification problem to a simple rule (**x > 0**).

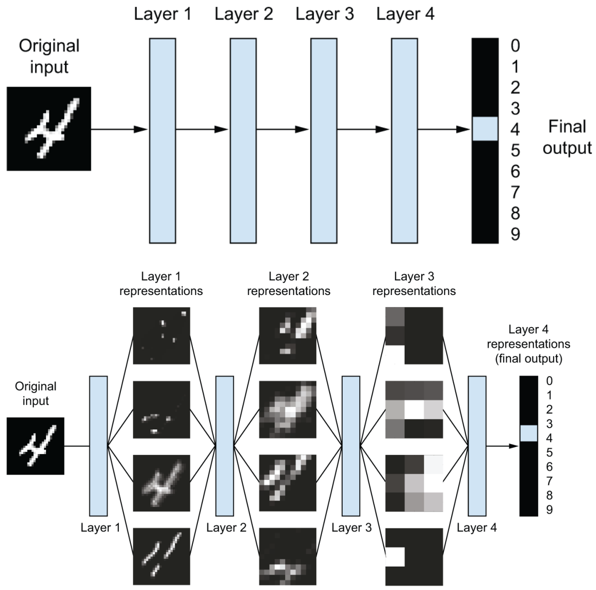
Deep Learning

- Deep Learning is a specific subfield of Machine Learning.
- A new take on learning representations from data, which emphasizes learning successive layers of increasingly meaningful representations.
- The *deep* in "Deep Learning" stands for this idea of successive layers of representations.
  - Modern deep learning often involves tens or even hundreds of successive layers of representations, and they are all learned automatically from exposure to training data.
- The layered representations are learned via models called neural networks, structured in literal layers stacked on top of each other.
- The term neural network is a reference to neurobiology.
  - Some of the central concepts in deep learning were developed in part by drawing inspiration from our understanding of the brain.
  - Deep learning models **are not models of the brain**.
- Other approaches to machine learning tend to focus on learning only one or two layers of representations of the data. (*shallow learning*)

Representations Learned in Deep Learning

- The network transforms the digit image into representations that are increasingly different from the original image and increasingly informative about the final result.
- Think of a deep network as a multistage information-distillation process, where information goes through successive filters and comes out increasingly purified.
- Deep Learning is technically a multistage way to learn data representations.
- It's a simple idea, but as it turns out, very simple mechanisms, sufficiently scaled, can end up looking like magic.

Transformation of an image of a digit by a network to recognize what digit it is.



Specific visual appearance is discarded while the core features determining the digit are purified at each layer.

Classical Machine Learning vs Deep Learning

Classical ML:

Raw Data

↓ Manual feature engineering

↓ Classifier / regressor

→ Output

DL:

Raw Data

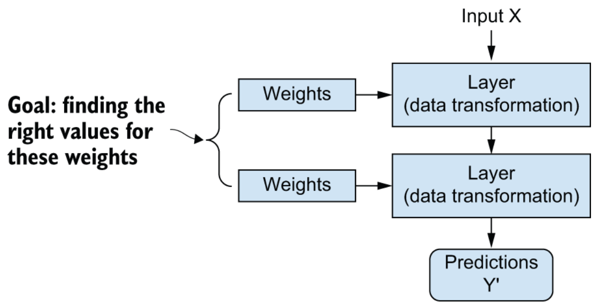
↓ Learned feature extraction (layers)

↓ Learned classifier

→ Output

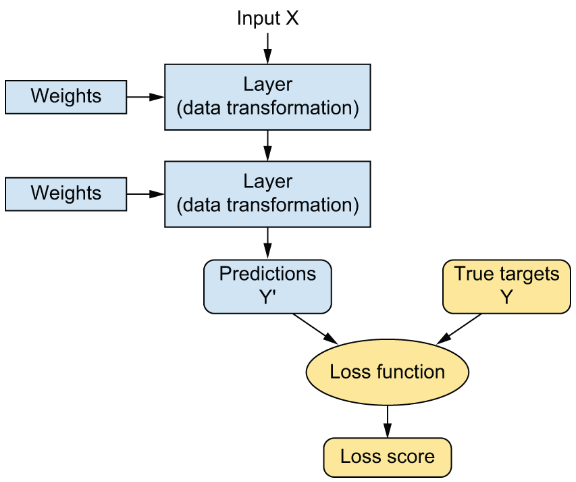
Deep Learning

- The specification of what a layer does to its input data is stored in the layer's weights, which in essence are a bunch of numbers.
- The transformation implemented by a layer is parameterized by its weights.
  - Weights are also sometimes called the parameters of a layer.
- In this context, learning means finding a set of values for the weights of all layers in a network, such that the network will correctly map example inputs to their associated targets.
- A deep neural network can contain tens of millions of parameters.
  - Finding the correct value for all of them may seem like a daunting task.
  - Modifying the value of one parameter will affect the behavior of all the others.



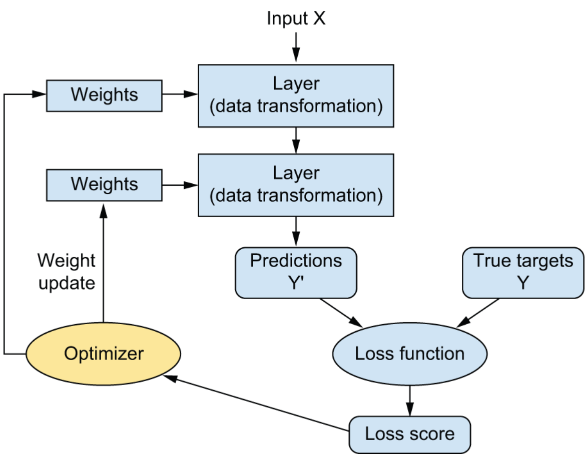
Deep Learning

- To control something, first you need to be able to observe it.
- To control the output of a neural network, you need to be able to measure how far this output is from what you expected.
  - This is the job of the loss function of the network, also sometimes called the objective function or cost function.
- The loss function takes the predictions of the network and the true target (what you wanted the network to output) and computes a distance score,
  - A distance score captures how well the network has done on this specific example.



Deep Learning

- The fundamental trick in deep learning is to use this score as a feedback signal to adjust the value of the weights a little, in a direction that will lower the loss score for the current example.
- This adjustment is the job of the optimizer, which implements what's called the Backpropagation algorithm: the central algorithm in deep learning.
- Initially, the weights of the network are assigned random values, so the network merely implements a series of random transformations.
    - Naturally, its output is far from what it should ideally be, and the loss score is accordingly very high.
    - With every example the network processes, the weights are adjusted a little in the correct direction, and the loss score decreases.



The optimizer closes the training loop.

Deep Learning

- This is the training loop, which, repeated a sufficient number of times (typically tens of passes over thousands of examples), yields weight values that minimize the loss function.

- **A trained network**
    - A network with a minimal loss for which the outputs are as close as they can be to the targets.

What makes deep learning different?

Deep learning has several properties that justify its status as an AI revolution.

- **Simplicity**
  - Deep learning makes problem solving much easier, because it automates what used to be the most crucial step in a machine learning workflow: feature engineering.
- **Scalability**
  - Deep learning is highly amenable to parallelization on GPUs or more specialized machine learning hardware, so it can take full advantage of Moore's law.
  - In addition, deep learning models are trained by iterating over small batches of data, allowing them to be trained on datasets of arbitrary size.
- **Versatility and reusability**
  - Unlike many prior machine learning approaches, deep learning models can be trained on additional data without restarting from scratch, making them viable for continuous online learning, an important property for very large production models.
  - Trained deep learning models are repurposable and thus reusable: this is the big idea behind "foundation models"

We may not be using neural networks many decades from now, but whatever we use will directly inherit from modern deep learning and its core concepts.

Generative AI

- Generative AI is powered by very large foundation models that learn to reconstruct the text and image content fed into them
  - reconstruct a sharp image from a noisy version.
  - predict the next word in a sentence.
- In self-supervised learning the targets are taken from the input itself.
  - It enables models to use vast amounts of unlabeled data.
  - Manual data annotations had bottlenecked previous brands of machine learning.
- **Foundation Models** are large models trained on humongous amounts of data, which can be used across many new tasks with little retraining, or even none at all.
  - Some of these foundation models have hundreds of billions of parameters and are trained on over 1 petabyte of data, at the cost of tens of millions of dollars.
  - These foundation models have already memorized so much and can solve new problems merely via prompting
- Generative AI only rose to mainstream awareness in 2022, but it has a long history.
  - The earliest experiments with text generation date back to the 1990s.

Summary

- We looked at the AI, ML, and DL landscape and defined the hierarchy and relationship between Artificial Intelligence, Machine Learning, and Deep Learning.
- We looked at learning representations and understood how deep learning models transform raw data into increasingly meaningful representations.
- We looked at the mechanics of the training loop and learned how weights, loss functions, and optimizers work together to allow a model to learn from data.
- We discussed the modern era and the revolutionary impact of scale, foundation models, and the rise of Generative AI.

References

- The content and figures used in these slides are extracted from the textbook **Deep Learning with Python, Third Edition - François Chollet**.

Next Session:
Mathematical Building Blocks in Deep Learning

Any Questions?